Consolidated Literature Review

Index	Source	Key Points	Topic Covered	Link
1	Title: Machine learning in medicine: a practical introduction to natural language processing Journal: BMC Medical Research Methodology	• **Preprocessing matters:** Cleaning and transforming free text into structured numeric features is a crucial first step before any model (including a difficulty predictor) can learn effectively. • **Feature extraction can be thematic:** Topic models like LDA help summarize text into themes or patterns — these could be linked to levels of report difficulty or complexity. • **Supervised classification pipelines apply:** The paper's example shows that standard ML classifiers can learn from textual features to predict labels; this pipeline can be adapted for our project once we define difficulty labels.	Bac…	https://link.springer.com/article/10.1186/s12874-021-01347-1
2	Title: An End-to-End Natural Language Processing Application for Prediction of Medical Case Coding Complexity: Algorithm Development and Validation Journal: JMIR Medical Informatics (Journal of Medical Internet Research)	• **Clinical free text can be used to predict a "difficulty/complexity" label:** The study showed that information contained in discharge notes and reports is sufficient to estimate how complex a case will be, without needing the final outcome labels. This supports the idea that text alone can act as a proxy signal for how hard a downstream model task may be. • **Simple text representations are effective starting points:** Traditional features like TF-IDF and basic embeddings performed competitively with larger transformer models, especially when data was limited. This suggests our group can begin with lightweight NLP features to score report difficulty before moving to more complex or synthetic-data approaches. • **Difficulty labels are inherently noisy and imbalanced:** Even expert human coders agreed only about two-thirds of the time, and multiclass prediction was challenging. Our project should expect similar subjectivity and may need strategies like synthetic data augmentation or coarser "easy vs hard" groupings.	Bac…	https://medinform.jmir.org/2023/1/e38150

3	Title: Use of unstructured text in prognostic clinical prediction models: a systematic review Journal: Journal of the American Medical Informatics Association (JAMIA)	• **Text data adds valuable information beyond structured data:** Across 126 studies, models that combined unstructured text with structured data consistently outperformed models using only one data type, suggesting that clinical free text provides important signals that improve prediction quality. This supports our goal of leveraging free text as a meaningful feature source for estimating difficulty before running vision models. • **A range of text representations and ML methods are effective:** Both traditional sparse representations (e.g., TF-IDF) and dense embeddings were used successfully with various machine learning approaches—from logistic regression to neural networks. This means our project has flexibility in choosing NLP features and models for capturing text characteristics that correlate with difficulty. • **External validation and explainability are under-addressed:** Very few models in the literature were externally validated or had strong interpretability, highlighting a common research gap. For our project, this points to an opportunity to focus on robust validation of our difficulty estimates and clear explanation of how text patterns relate to predicted difficulty.	Bac…	https://academic.oup.com/jamia/article/29/7/1292/6574714
4	Natural language processing in medicine: A review	• Natural language processing (NLP) is a form of machine learning which enables the processing and analysis of free text. • NLP has the ability to improve time-efficiencies and personalise healthcare through its strong predictive capabilities • Data must be pre-processed to ensure minimal loss within the model. This often involves NLP subtasks based on syntactic and semantic analysis such as tokenisation, lemmatisation and word sense disambiguation.	NLP	https://www.sciencedirect.com/science/article/pii/S2210844021000411
5	MIMIC-CXR & MIMIC-CXR-JPG Database	• All images have been de-identified to protect patient privacy. • Contains 377,110 chest radiographs with free-text radiology reports. • MIMIC-CXR-JPG is derived entirely from the MIMIC-CXR	Dataset	https://physionet.org/content/mimic-cxr/2.1.0/ https://arxiv.org/abs/1901.07042

		database, and aims to provide a convenient processed version of MIMIC-CXR		
6	An open dataset of chest x-rays with radiologist's annotations	• Notable public datasets of CXR include ChestX-ray8, ChestX-ray148, Padchest9, CheXpert2, and MIMIC-CXR10. ChestX-ray14	Dataset	https://pmc.ncbi.nlm.nih.gov/articles/PMC9300612/
7	Computer vision to enhance healthcare domain: An overview of features, implementation, and opportunities	• Aim: Discuss the significant applications of CV in healthcare • Computer vision in healthcare focuses on the interpretation of images and videos. • Some applications: ○ CVs have been used in many healthcare applications to assist medical staff in treating patients more efficiently. ○ CV development in healthcare in recent years has made diagnostics quicker and more precise. ○ ML models are integrated into medical procedures to increase surgical success rates and lower risk. ○ CVs may also be utilised to monitor the behaviour of hospital staff members and visitors. ○ CV systems to reduce false positives during the diagnostic process. ○ A CV may help reduce the time wasted on pointless diagnostic tests and provide medical professionals with the tools they need to order more precise diagnoses and successful treatments. ○ CV technology helps doctors respond swiftly by enabling the identification of symptoms even before they become evident. ○ CV is successful because it can see patterns that help physicians diagnose ailments early. ○ CV also reduces the possibility of human error in the healthcare industry. ○ One of the most crucial uses for CV is tumour detection	CV …	https://www.sciencedirect.com/science/article/pii/S2949866X24000662
8	Clinical-BERT: Vision-language pre-training for radiograph diagnosis	• Aim: Propose and evaluate a vision-language pre-training model, Clinical BERT, for the medical domain	NLP	https://ojs.aaai.org/index.php/AAAI/article/download/2

	and reports generation	• Clinical BERT is trained via 4 tasks ○ (1) Clinical diagnosis: predict disease labels from radiographs ○ (2) Masked Language Modelling (MLM): predict missing words in reports ○ Masked MeSH Modelling (MMM): focus on predicting MeSH terms ○ Image-MeSH Matching (IMM): learn alignments between image regions and medical terms in reports • Results: ○ Vision-language pre-training improves both radiograph diagnosis and report generation tasks compared to single-modality models ○ Achieves state-of-the-art performance on multiple benchmark datasets ○ Uncertainty handling: Model learns to recognize ambiguous cases through exposure to uncertain labels (-1) from CheXpert • Implications: ○ Pre-trained medical language models (specialized BERT) significantly outperform general-purpose language models for clinical text ○ Automated label extraction (CheXpert labeler) is reliable enough for large-scale training without manual annotation ○ Multi-task learning improves generalization across different medical imaging tasks		0204/19963
9	An Open-Source Reproduction and Enhancement of CheXNet for Chest X-ray Disease Classification	• Aim: ○ Replicate the original CheXNet model as closely as possible ○ Evaluate and improve performance metrics such as AUC-ROC and F1 scores across all 14 disease classes ○ Explore whether newer deep learning techniques, particularly Vision Transformers (ViTs), could offer performance improvements over traditional convolutional neural networks	CV …	https://arxiv.org/html/2505.06646v2

		• Results: ◦ CheXNet: This model established a performance baseline for our project, achieving reasonable AUCs but low F1 scores, particularly on underrepresented classes like pneumonia. ▪ Simply replicating CheXNet's structure was not sufficient to achieve high F1 scores, especially for underrepresented diseases. ◦ DACNet: Achieved a strong balance between interpretability and predictive power, reaching an average AUC of 0.85 and an F1 score of 0.39 ◦ ViTs: This model produced an average test AUC of 0.794, a test loss of 0.159, and an average test F1 score of 0.111 across all diseases. ▪ This model did not beat CheXNet's AUC score for any of the 14 diseases		
10	Multimodal vision-language models in chest x-ray analysis: a study of generalization, supervision, and robustness	• Aim: Systematic evaluation of six leading unimodal and multimodal models for chest X-ray (CXR) classification using four widely adopted datasets. • Multimodal Models (CheXzero, CXR-LLaVA): ◦ Strengths: ▪ Excellent at zero-shot tasks (no training needed) ▪ Generalize well to new/different datasets ▪ Understand semantic meaning across images and text ◦ Weaknesses: ▪ Performance drops after fine-tuning on specific tasks ▪ Prone to overfitting with noisy or cross-lingual data • Unimodal Models (Single-modality: image OR text only): ◦ Strengths: ▪ Improve significantly with supervised training ▪ Highly adaptable when given labeled data ◦ Weaknesses: ▪ Struggle when tested on different datasets (domain	CV …	https://link.springer.com/article/10.1007/s13534-025-00535-y

		shift) ■ Need good quality training data to perform well ● TorchXRayVision: ○ Strengths: ■ Consistent performance across different datasets ■ Lightweight and practical for limited resources ■ Decent generalization without heavy requirements ○ Weaknesses: ■ Lower overall accuracy compared to newer models ■ Not state-of-the-art, but reliable baseline		
11	ClinicalBERT: Modeling Notes and Predicting Hospital Readmission	● Aim: ○ Adapt BERT to clinical domain ○ Improve representation of clinical free text ○ Enhance performance on downstream medical prediction tasks ○ Demonstrate effectiveness via readmission prediction using MIMIC-III ● Strengths ○ Domain specific language understanding ■ Clinical pretraining improves performance over general BERT → implication is better handling of terminology, abbreviations, and uncertainty phrases ○ Contextual embedding ■ Outperforms Word2Vec, bag-of-words, and BiLSTM in downstream tasks → implication is it captures subtle differences that may correlate with CV model difficulty ○ Strong performance in supervised classification ■ Achieved higher AUROC in readmission prediction compared to baselines → implication is fine-tuning for Easy/Moderate/Hard classification is methodlogically validated ○ Reduced manual feature engineering ■ Transformer-based representations outperform manual text features → implication is there's no	NLP	https://arxiv.org/pdf/1904.05342

		need for handcrafted severity rules or keyword lists; embedding encode latent signals automatically ● Weakenesses ○ Trained on general clinical notes (not radiology specific) ■ Pretrained on MIMIC-III notes, not eluviely imaging reports → implication is it amy not fully optimize for radiology stle structure without further fine tuning ○ Computationally intensive ■ Large transformer architecture → requires GPU, longer training time, higher memory usage compared to TF-IDF or classical ML ○ Requires labeled data for fine tuning ■ Performance gains show after supervised fine-tuning → implication is your difficulty labels must be well defined and sufficiently large for effective training ○ Interpretability is limited ■ Attention mechanism provide partial insight, but model remains largely black-box → implication is explaining why a case is "hard" may require additional interpretability tools		
12	A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis	● Comparison of the performance between health-care professionals in these 14 studies, when restricting the analysis to the contingency table for each study reporting the highest accuracy, found a pooled sensitivity of 87·0% (95% CI 83·0–90·2) for deep learning models and 86·4% (79·9–91·0) for health-care professionals, and a pooled specificity of 92·5% (95% CI 85·1–96·4) for deep learning models and 90·5% (80·6–95·7) for health-care professionals.	CV …	Liu X, Faes L, Kale A et al. A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis The Lancet Digital Health, 2019; 1, e271-e297
13	Diagnostic precision of a deep		CV …	Güzel, H. E., Aşcı, G.,

	learning algorithm for the classification of non-contrast brain CT reports			Demirbilek, O., Özdemir, T. D., & Erekli, P. B. (2025). Diagnostic precision of a deep learning algorithm for the classification of non-contrast brain CT reports. *Frontiers in radiology*, *5*, 1509377. https://doi.org/10.3389/fradi.2025.1509377
14	Investigation of Radiologist Diagnostic Difficulty Prediction Without CT Images		CV …	
15	Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges		CV …	Poon, E. G., Lemak, C. H., Rojas, J. C., Guptill, J., & Classen, D. (2025). Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges. *Journal of the American Medical Informatics Association : JAMIA*, *32*(7), 1093–1100. https://doi.org/10.1093/jamia/ocaf065